Data Science Capstone Project

EEG-Based Mental Stress Detection

Final Report

Course:	Data Science Capstone Project
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Abstract

This report presents a data science project for EEG-based mental stress detection. The aim of the project is to classify EEG recordings into stress and non-stress states using machine learning. The dataset consists of EEG recordings collected under several experimental conditions, including complex mathematical problem solving, Trier mental challenge testing, Stroop colour word testing, horror video stimulation, and relaxing music listening. The stress-inducing conditions were grouped into the stress class, while the relaxing music condition was used as the non-stress class.

The methodology includes data loading, selection of relevant EEG channels, signal preprocessing, window-based segmentation, feature extraction, and model evaluation. Five EEG channels were used: AF3, T7, Pz, T8, and AF4. The EEG signals were filtered using a 0.5–45 Hz bandpass filter and divided into 10-second windows. For each window, statistical features and frequency-band power features were extracted. Several machine learning models were compared, including Logistic Regression, Random Forest, Support Vector Machine, K-Nearest Neighbors, and Gradient Boosting.

The results show that window-based classification improves the amount of available training data, but evaluation must be performed carefully because windows from the same recording can be highly similar. Therefore, GroupKFold validation was used to keep windows from the same EDF recording together. Under this stricter evaluation, Gradient Boosting achieved the best Macro F1-score. The final model performed well in detecting stress windows, but struggled with the minority non-stress class, mainly due to class imbalance. Overall, the project demonstrates that EEG features can be used for stress detection, while also highlighting the challenges of imbalanced biomedical signal classification.

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1 Introduction

1.1 Problem Statement

Mental stress is an important factor affecting health, cognitive performance, productivity, and general well-being. In many situations, stress is assessed through questionnaires or self-reports. Although these methods are easy to use, they are subjective and may not always reflect the actual physiological state of a person. Because of this, physiological signals such as electroencephalography (EEG) can provide a more objective way of analyzing stress-related changes.

This project focuses on building a machine learning classifier for mental stress detection using EEG recordings. The task is formulated as a binary classification problem, where EEG signal windows are classified as either stress or non-stress. The stress class includes recordings from mentally demanding or emotionally stimulating tasks, while the non-stress class includes relaxed-state recordings.

1.2 Objectives

The main objectives of this project are:

- to explore and understand the provided EEG dataset;
- to define stress and non-stress classes based on the experimental conditions;
- to preprocess EEG signals and select relevant EEG channels;
- to extract statistical and frequency-domain features from EEG windows;
- to train and compare several machine learning models;
- to evaluate the models using suitable metrics for imbalanced classification;
- to analyze the limitations and possible improvements of the final model.

1.3 Dataset Overview

The dataset used in this project was provided through a soft copy link from Mendeley Data. It contains EEG recordings stored as EDF files and organized by experimental condition. The available conditions include complex mathematical problem solving, horror video stimulation, Stroop colour word testing, Trier mental challenge testing, and relaxing music listening.

The selected EEG channels used in this project are AF3, T7, Pz, T8, and AF4. The recordings have a sampling frequency of 128 Hz. The four task-based conditions were grouped into the stress class, while the relaxing music condition was used as the non-stress class.

1.4 Background and Challenges

EEG data is a challenging type of biomedical signal because it can contain noise, artifacts, and strong individual variability between subjects. In addition, the dataset used in this project is imbalanced, with more stress recordings than non-stress recordings. This affects model training and evaluation because a model may achieve high accuracy by mainly predicting the majority class. For this reason, balanced accuracy and Macro F1-score are used in addition to standard accuracy.

2 Literature Review

3 Methodology

4 Results

4.1 Dataset Distribution

Table 1: Distribution of EEG recordings by condition.

Condition	Number of EDF Files	Class
Complex Mathematical Problem Solving (CMPS)	22	Stress
Horror Video Stimulation	22	Stress
Stroop Colour Word Test (SCWT)	24	Stress
Trier Mental Challenge Test (TMCT)	24	Stress
Relaxing Music	20	Non-stress

Table 2: Class distribution after 10-second window segmentation.

Class	Number of Windows	Percentage
Stress	3037	86.6
Non-stress	470	13.4

4.2 Exploratory Data Analysis

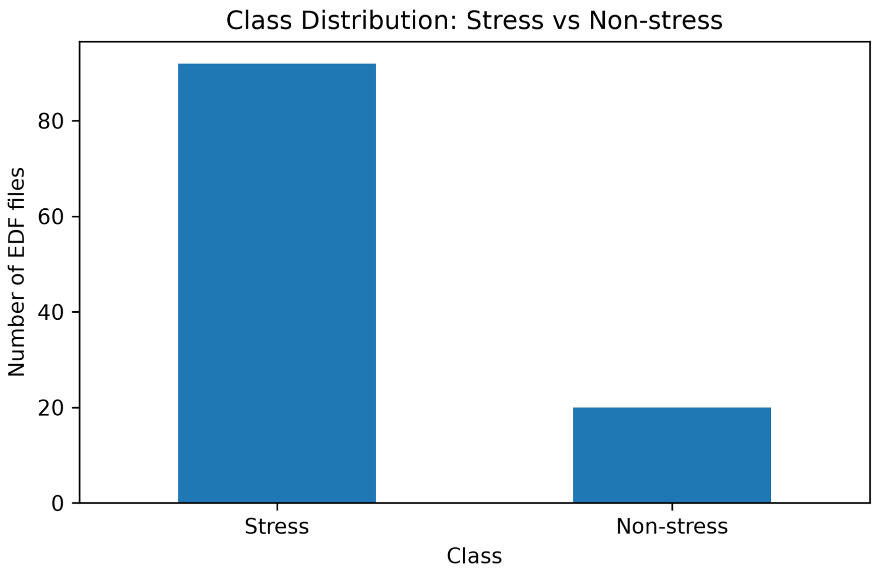

Figure 1: Class distribution of stress and non-stress EEG recordings.

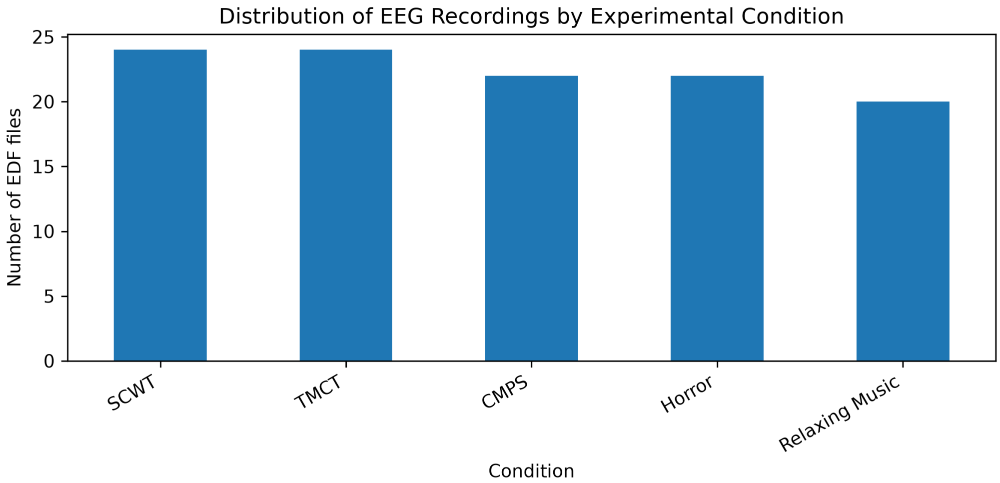

Figure 2: Distribution of EEG recordings by experimental condition.

4.3 Signal Visualization

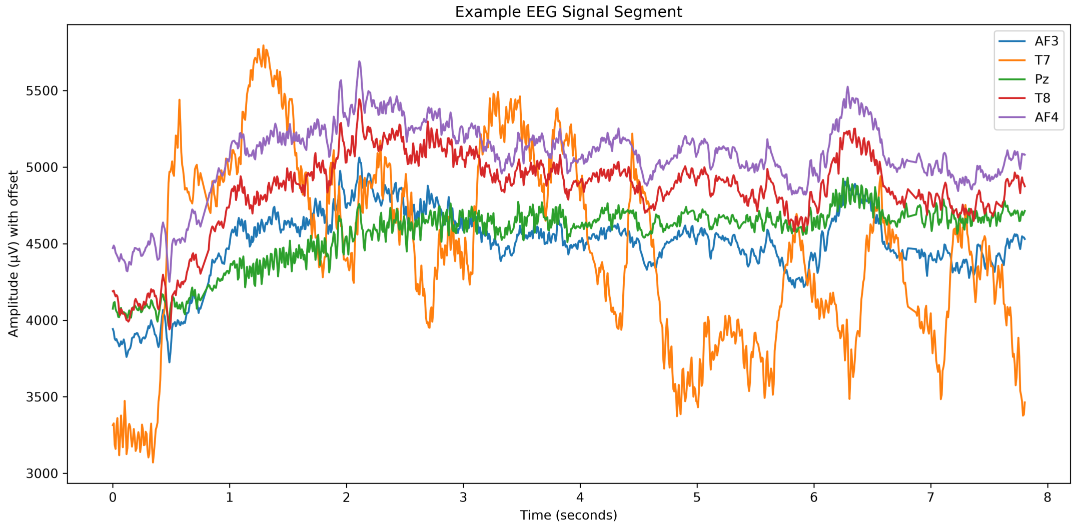

Figure 3: Example raw EEG signal segment from selected channels.

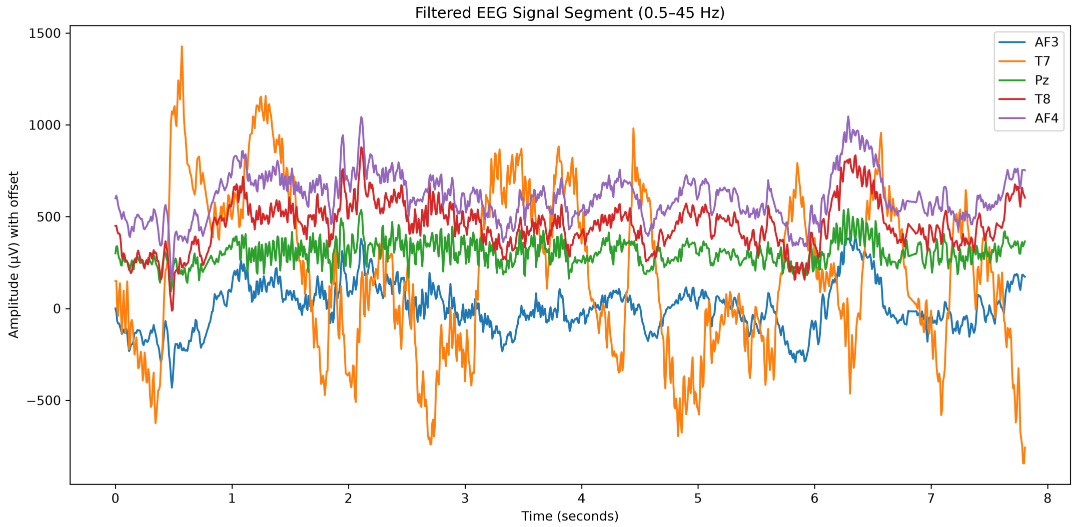

Figure 4: Filtered EEG signal segment after applying a 0.5–45 Hz bandpass filter.

4.4 Model Comparison

Table 3: Model comparison using GroupKFold validation by EDF file.

Model	Accuracy	Balanced Accuracy	Precision Macro	Recall Macro	F1
Gradient Boosting	0.8654	0.5809	0.6758	0.5809	0.
KNN	0.8349	0.5565	0.5865	0.5565	0.
SVM	0.7025	0.6431	0.5685	0.6431	0.
Random Forest	0.8389	0.5389	0.5884	0.5389	0.
Logistic Regression	0.6541	0.6200	0.5491	0.6200	0.

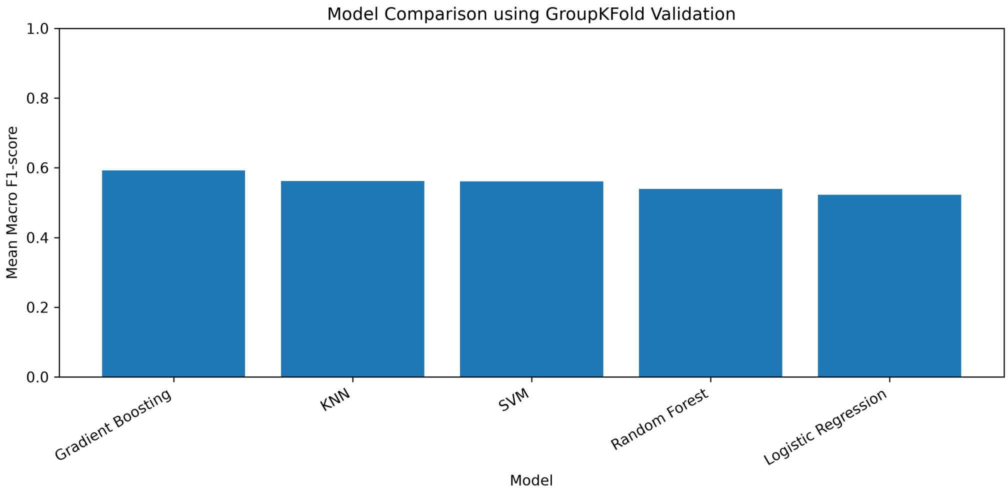

Figure 5: Model comparison using GroupKFold validation.

4.5 Final Model Performance

Table 4: Classification report for the final Gradient Boosting model.

Class	Precision	Recall	F1-score	Support
Non-stress	0.69	0.16	0.25	141
Stress	0.83	0.98	0.90	573
Accuracy	–	–	0.82	714
Macro Average	0.76	0.57	0.58	714
Weighted Average	0.80	0.82	0.77	714

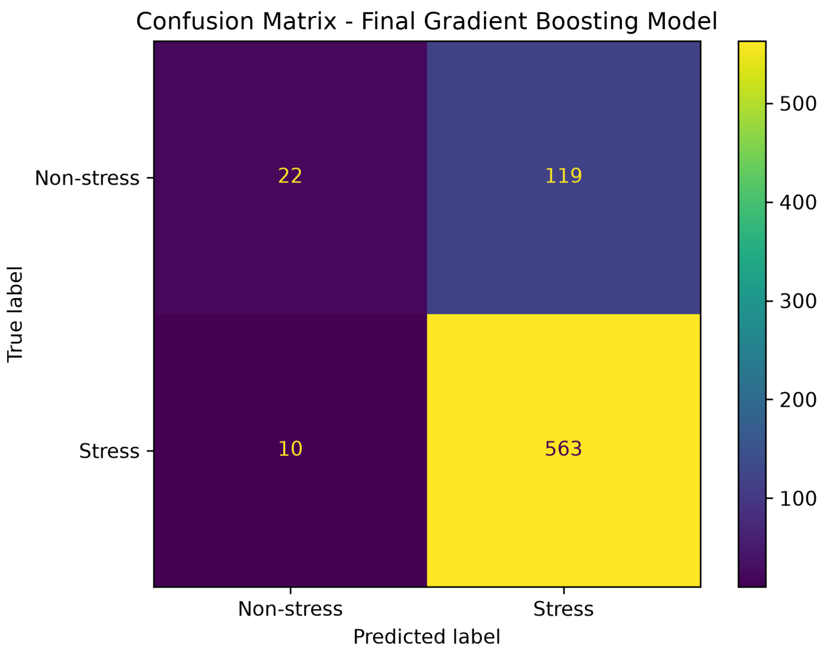

Figure 6: Confusion matrix of the final Gradient Boosting model.

4.6 Feature Importance

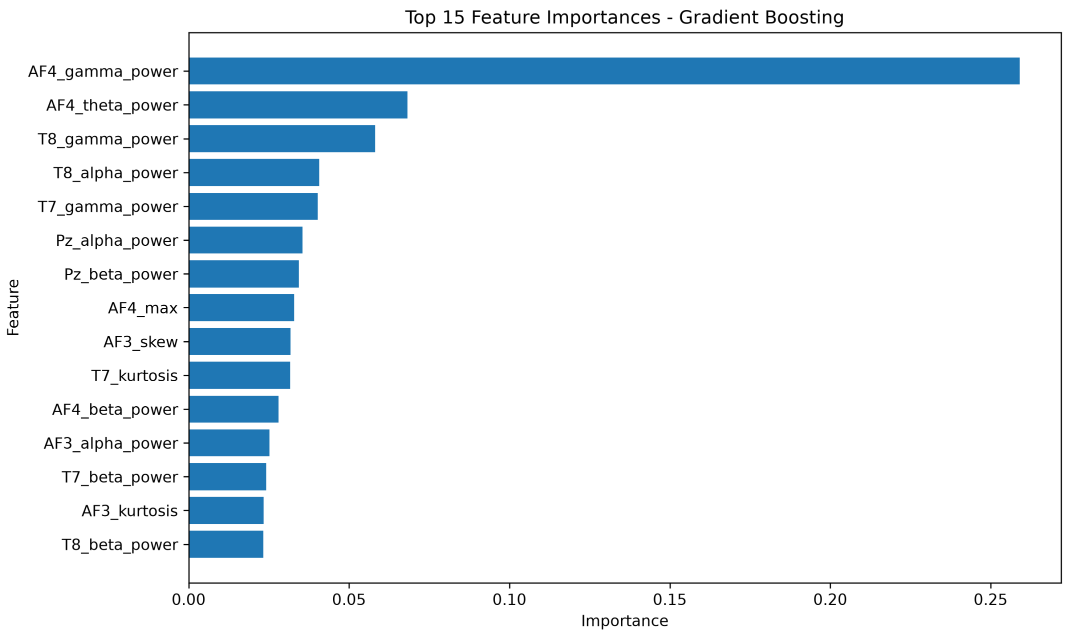

Figure 7: Top 15 feature importances of the final Gradient Boosting model.

5 Discussion

6 Conclusion

References

[1] Mendeley Data. *An EEG Recordings Dataset for Mental Stress Detection*. Available at: https://data.mendeley.com/datasets/wnshbvdxs2/1

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[3] Pedregosa, F. et al. *Scikit-learn: Machine Learning in Python*. Journal of Machine Learning Research, 2011.